

Cambridge International AS & A Level

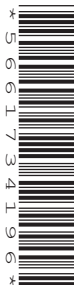
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FURTHER MATHEMATICS

9231/32

Paper 3 Further Mechanics

May/June 2022

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

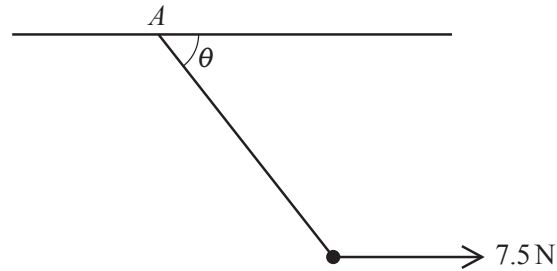
- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

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A particle of weight 10 N is attached to one end of a light elastic string. The other end of the string is attached to a fixed point A on a horizontal ceiling. A horizontal force of 7.5 N acts on the particle. In the equilibrium position, the string makes an angle θ with the ceiling (see diagram). The string has natural length 0.8 m and modulus of elasticity 50 N .

- (a) Find the tension in the string. [2]

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- (b)** Find the vertical distance between the particle and the ceiling. [3]

[illegible]

[5]

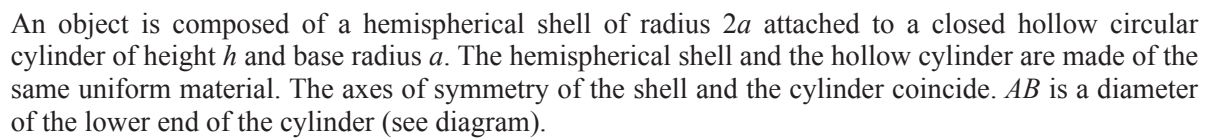
[illegible]

- 3** A particle P is moving in a horizontal straight line. Initially P is at the point O on the line and is moving with velocity 25 m s^{-1} . At time $t \text{ s}$ after passing through O , the acceleration of P is $\frac{4000}{(5t+4)^3} \text{ m s}^{-2}$ in the direction PO . The displacement of P from O at time t is $x \text{ m}$.

Find an expression for x in terms of t .

[5]

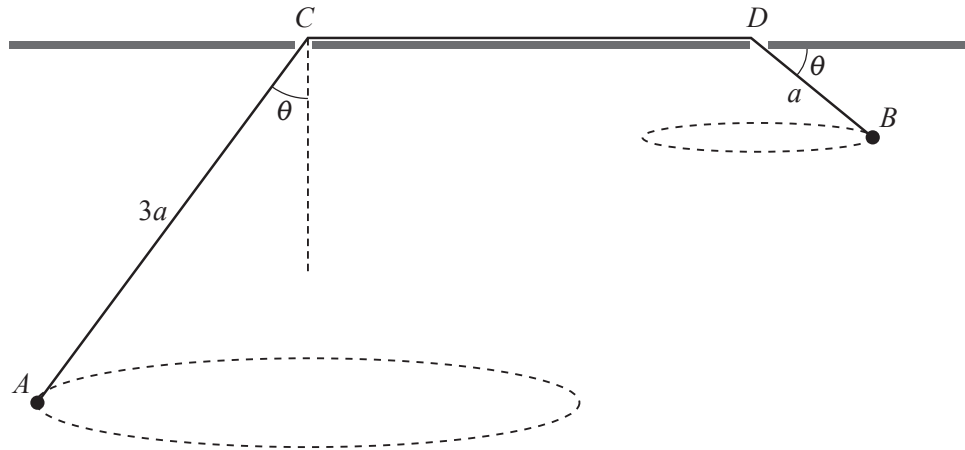
[illegible]



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- This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

The object is placed on a rough plane which is inclined to the horizontal at an angle θ , where $\tan \theta = \frac{2}{3}$. The object is in equilibrium with AB in contact with the plane and lying along a line of greatest slope of the plane.

- (b) Find the set of possible values of h , in terms of a . [4]



A light inextensible string AB passes through two small holes C and D in a smooth horizontal table where $AC = 3a$ and $DB = a$. A particle of mass m is attached at the end A and moves in a horizontal circle with angular velocity ω . A particle of mass $\frac{3}{4}m$ is attached to the end B and moves in a horizontal circle with angular velocity $k\omega$. AC makes an angle θ with the downward vertical and DB makes an angle θ with the horizontal (see diagram).

Find the value of k .

[7]

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

After the collision, the kinetic energy of A is equal to the kinetic energy of B .

- (b) Given that $\tan \alpha = \frac{2}{3}$, find the possible values of k . [5]

[illegible]

[illegible]

(b) Find the value of T .

[4]

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(c) Find the horizontal and vertical displacements of the particles from O when they collide.

[3]

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Additional page

If you use the following page to complete the answer to any question, the question number must be clearly shown.

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